Multiplying Numbers in Scientific Notation

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Scientific Notation

Scientific notation expresses a number in the form

$$a \times 10^n$$

where $1 \leq |a| < 10$ and n is an integer.

Multiplying

To **multiply numbers in scientific notation**:

multiply their **coefficients**

and **add** their **exponents**.

Product of Powers Rule

Use the **product of powers rule**:

$$10^m \times 10^n = 10^{m+n}$$

Example:

$$10^2 \times 10^3 = 10^{2+3} = 10^5$$

Why Does It Work?

$$10^2 \times 10^3$$

$$= (10 \times 10) \times (10 \times 10 \times 10)$$

$$= 10^5$$

Five 10s multiplied together.

The General Rule

$$(a \times 10^m)(b \times 10^n)$$

$$= (a \times b) \times 10^{m+n}$$

Remember: Multiply, Add, Normalize!

Multiply the coefficients.

Add the exponents.

Normalize the result.

Proper Scientific Notation

If the product of the coefficients is not between 1 and 10, rewrite the result in **proper scientific notation**.

$$30 \times 10^5 = 3 \times 10^6$$

Final Check

Remember to check that the final coefficient satisfies

$$1 \leq |a| < 10$$

Examples

Example 1

Multiply.

$$(2 \times 10^3)(3 \times 10^2)$$

$$= (2 \times 3) \times 10^{3+2}$$

$$= 6 \times 10^5$$

Example 2

Multiply.

$$(1.5 \times 10^4)(2 \times 10^3)$$

$$= (1.5 \times 2) \times 10^{4+3}$$

$$= 3 \times 10^7$$

Example 3

Multiply.

$$(5 \times 10^3)(6 \times 10^2)$$

Multiply and add:

$$= 30 \times 10^5$$

Example 3 (continued)

Normalize:

$$30 \times 10^5 = 3 \times 10^1 \times 10^5$$

$$= 3 \times 10^6$$

Example 4

Multiply.

$$(3 \times 10^{-4})(2 \times 10^{-2})$$

$$= (3 \times 2) \times 10^{-4+(-2)}$$

$$= 6 \times 10^{-6}$$

Example 5

Multiply.

$$(4 \times 10^{-3})(2.5 \times 10^{5})$$

Multiply and add:

$$= 10 \times 10^{2}$$

Example 5 (continued)

Normalize:

$$10 \times 10^2 = 1 \times 10^1 \times 10^2$$

$$= 1 \times 10^3$$

Practice Exercises

Practice 1

Multiply each pair of numbers expressed in scientific notation.

$$(3 \times 10^4)(2 \times 10^3)$$

Practice 2

Multiply each pair of numbers expressed in scientific notation.

$$(5 \times 10^6)(4 \times 10^2)$$

Practice 3

Multiply each pair of numbers expressed in scientific notation.

$$(1.2 \times 10^5)(3 \times 10^4)$$

Practice 4

Multiply each pair of numbers expressed in scientific notation.

$$(6 \times 10^{-3})(2 \times 10^{5})$$

Practice 5

Multiply each pair of numbers expressed in scientific notation.

$$(4.5 \times 10^{-2})(2 \times 10^{-4})$$

Practice 6

Multiply each pair of numbers expressed in scientific notation.

$$(7 \times 10^3)(8 \times 10^{-5})$$

Practice 7

Multiply each pair of numbers expressed in scientific notation.

$$(2.4 \times 10^{7})(5 \times 10^{-3})$$

Practice 8

Multiply each pair of numbers expressed in scientific notation.

$$(3.6 \times 10^{-6})(4 \times 10^{-2})$$

Practice 9

Multiply each pair of numbers expressed in scientific notation.

$$(8.5 \times 10^4)(3 \times 10^6)$$

Practice 10

Multiply each pair of numbers expressed in scientific notation.

$$(9.2 \times 10^{-5})(6 \times 10^{-3})$$

Activities

Activity 1

Multiply each pair of numbers expressed in scientific notation.

$$(4 \times 10^5)(3 \times 10^2)$$

Activity 2

Multiply each pair of numbers expressed in scientific notation.

$$(6 \times 10^4)(5 \times 10^3)$$

Activity 3

Multiply each pair of numbers expressed in scientific notation.

$$(2.5 \times 10^6)(4 \times 10^3)$$

Activity 4

Multiply each pair of numbers expressed in scientific notation.

$$(8 \times 10^{-4})(3 \times 10^{6})$$

Activity 5

Multiply each pair of numbers expressed in scientific notation.

$$(3.5 \times 10^{-3})(4 \times 10^{-5})$$

Activity 6

Multiply each pair of numbers expressed in scientific notation.

$$(6 \times 10^2)(7 \times 10^{-6})$$

Activity 7

Multiply each pair of numbers expressed in scientific notation.

$$(3.2 \times 10^{8})(4 \times 10^{-2})$$

Activity 8

Multiply each pair of numbers expressed in scientific notation.

$$(2.5 \times 10^{-7})(6 \times 10^{-3})$$

Activity 9

Multiply each pair of numbers expressed in scientific notation.

$$(7.5 \times 10^3)(4 \times 10^5)$$

Activity 10

Multiply each pair of numbers expressed in scientific notation.

$$(8.4 \times 10^{-4})(5 \times 10^{-2})$$

Thank you!